

PROGRAM – 5

Demonstrate CI Integration by Running Jenkins in a Docker Container

Project Name: program_devops_5

Technologies Used: Docker, Jenkins, Git

System Requirements

Hardware Requirements

- Minimum 4 GB RAM
- Minimum 20 GB free disk space
- Processor: Dual Core or higher

Software Requirements

- Ubuntu Linux / Windows / macOS
- Docker Engine installed
- Git installed
- Web browser (Chrome / Firefox)

Architecture / Workflow

Developer → GitHub → Jenkins (Docker) → Docker Image Build

Step 1 – Pull Jenkins Docker Image

`docker pull jenkins/jenkins:lts`

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program5_devops
meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program5_devops
meghana@meghana-HP-Laptop-15s-dr3xxx: $ cd program5_devops
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program5_devops$ docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
13cc39f8244a: Already exists
5687d4df0728: Pull complete
2f1ecb09fa36: Pull complete
68cdef480092: Pull complete
e0e73f34dee9: Pull complete
687044b042c5: Pull complete
581deb283b89: Pull complete
a9230f7c9c31: Pull complete
d7dd2e630238: Pull complete
3fb70446889e: Pull complete
986be969b0a: Pull complete
d2519f76d4c6: Pull complete
Digest: sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program5_devops$
```

Step 2 – Create Jenkins Volume

`docker volume create jenkins_data`

Step 3 – Run Jenkins Container

`docker run -d \ --name jenkins \ -p 8080:8080 \ -p 50000:50000 \ -v jenkins_data:/var/jenkins_home \ jenkins/jenkins:lts`

```
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meghana@meghana-HP-Laptop-15s-dr3xxx: $ mkdir program5_devops
meghana@meghana-HP-Laptop-15s-dr3xxx: $ cd program5_devops
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program5_devops$ docker pull jenkins/jenkins:lts
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13cc39f8244a: Already exists
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a9230f7c9c31: Pull complete
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Digest: sha256:1e407df1a9bcb122cc4d71d4ce44d2e3b2be31af004c157453a939f13ae691c7
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program5_devops$ docker run -d \
--name jenkins \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_data:/var/jenkins_home \
jenkins/jenkins:lts
92c6ceac2d84e1ca0af4daf17d34378c126a546e4c4bf98e042d13dee09ad85
docker: Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint jenkins (bdc97ced7927a5899461d253799a06a5a042b3a51f22c21b1691c576bf1fb075): failed to bind host port 0.0.0.0:8080/tcp: address already in use

Run 'docker run --help' for more information
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program5_devops$
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program5_devops$
```

Step 4 – Access Jenkins UI

`http://localhost:8080`

